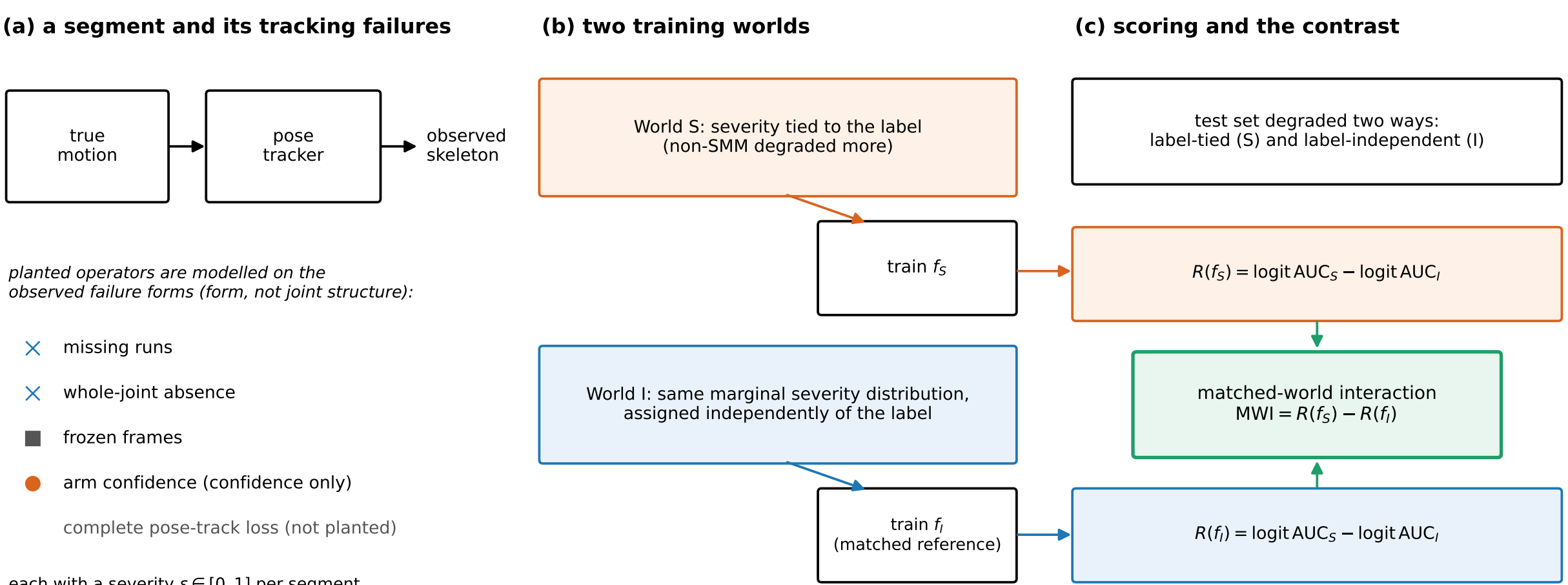
(a) a segment and its tracking failures
true motion
pose tracker
observed skeleton
planted operators are modelled on the observed failure forms (form, not joint structure):
missing runs
whole-joint absence
frozen frames
arm confidence (confidence only)
complete pose-track loss (not planted)
each with a severity $\varepsilon \in [0, 1]$ per segment
(b) two training worlds
World S: severity tied to the label (non-SMM degraded more)
train f_S
World I: same marginal severity distribution, assigned independently of the label
train f_I (matched reference)
(c) scoring and the contrast
test set degraded two ways: label-tied (S) and label-independent (I)
$R(f_S) = \text{logit AUC}_S - \text{logit AUC}_I$
matched-world interaction
$\text{MWI} = R(f_S) - R(f_I)$
$R(f_I) = \text{logit AUC}_S - \text{logit AUC}_I$
paired identifier-clustered resamples; a relative contrast, not a reliance score